ModbusControl – Release notes

v3.0.2 – 2026-10-08

Features

- Redesigned auto-scaling to use true per-monitor Windows DPI scaling.
- The *Custom Modbus Queries* window now supports zooming.

Changes

- Migrated to .NET (CORE) 9 because .NET Framework is no longer being developed. This also enables correct Windows scaling behaviour. The application is self-contained, so no .NET runtime needs to be installed.
- .NET 9 requires Windows 10 (1607+) or newer, so this drops support for Windows 7/8/8.1.
- The installer now displays the license text (GPLv3).

Fixes

- The redesigned auto-scaling fixes blurry text, overlapping text and mis-sized UI controls on multi-monitor setups.
- New windows now open centered on the screen, because centering on the parent window is unreliable in .NET 9.
- Project Configuration window: fixed the warning panel for command-line overrides, which wasn't displaying. Improved the override-warning and register numbering scheme tooltips.
- Improved texts and tooltips in the *Save Project File* and *Options* windows.
- The *About* window now shows the main window's DPI with its size, credits the creator/maintainer, and links to the LICENSE file (GPLv3).

v2.0.24 – 2026-05-07

Features

- The Find/Filter function now supports *regular expressions*, allowing more flexible search and filter patterns.
- When an mbc project file fails to load, the program now suggests that a newer version of the software may be required.

Fixes

- Save mbc project file: Resolved an exception occurring during save operations involving "*Project configuration settings"*.
- Load mbc project file: Long error messages during mbc project file loading are no longer truncated.
- Load mbc project file: Loading an mbc project file with an invalid string encoding now correctly throws an error instead of failing silently.
- Load mbc project file: Now correctly rejects 5-digit register numbering (40001–49999) when the project uses a 6-digit numbering scheme (e.g., 400001 or 400000), as originally intended.
- Project configuration screen: now displays the register numbering scheme more clearly, such as *[40001-49999] or [40001-49999 + 400001-465536]*.

- Auto-scaling: Improved auto-scaling behaviour by aligning with the latest recommended configuration practices.

v2.0.23 – 2026-03-10

Fixes

- Settings files: Moved from "*AppData\Roaming\E.D.&A\ModbusControl*" to "*AppData\Roaming\ModbusControl*", and from "*ProgramData\E.D.&A\ModbusControl*" to "*ProgramData\ModbusControl*". The files are migrated automatically.
- Mbc validation XSD file: The path stored in the settings is now automatically migrated to the new "*C:\Program Files\ModbusControl*" location. The Options window now has a button to revert the path to the default location.
- Mbc project file: loading the stringencoding name is now case-insensitive.
- Logging: Refactored the Logging screen to remove its dependency on an external component. Logging of reads and writes is now enabled by default.
- Diagnostics counters window: Counts are now right-aligned.

v2.0.22 – 2026-02-27

Fixes

- Pressing **Ctrl + C** in the grid after selecting a cell in the *Value* column now copies only the value string to the clipboard instead of all row data. The same applies when pressing **Ctrl + C** while editing a value.
- When an invalid value is entered in the grid, the error message now remains visible longer and does not disappear on the next poll.
- Installer: now installs this application in the "*C:\Program Files\ModbusControl*" directory instead of in "*C:\Program Files (x86)\E.D.&A\ModbusControl*", reflecting that the application is a 64-bit process.
- Updater: Resolved an issue with Admin-by-Request that prevented the Updater process from stopping before the update, which resulted in a warning that the Updater file was still in use during the update.

v2.0.21 – 2026-02-13

Features

- Added support for string format (up to 150 registers or 300 bytes) with multiple encodings.
- ModbusExport: Now supports strings and includes a *Continue on error* option.
- ModbusImport: Now supports strings and includes a *Continue on error* option.

Fixes

- The XSD validation schema now accepts uint32 enum values from 2,147,483,648 to 4,294,967,295.
- ModbusExport: You can now paste a filename or path into the filename field.
- ModbusImport: You can now paste a filename or path into the filename field.

v2.0.20 – 2025-10-11

Fixes

- Editing a register or bitfield value in the grid sometimes caused the old value to be written to the target instead of the new one.

v2.0.19 – 2025-09-24

Fixes

- Changing a register value by selecting a new value from the enumeration list no longer worked; this has been corrected.

v2.0.18 – 2025-09-09

Features

- Added a new [Custom Modbus Queries] window (in the *Extra* menu), replacing the old [Send Read], [Send Write], and [Send Custom Command] windows.
- Added saving and restoring of main-grid column widths, based on relative weight.
- Moved [Diagnostics Counters] and [Logging] to the [Extra] menu.
- Removed the [Import-Compare] feature.

Fixes

- In some cases, column widths unexpectedly reverted to earlier configurations.
- When using *Auto-start edit mode on typing*, the first digit was lost when editing non-enum values.
- Export window settings were not being saved or restored.
- Major refactoring of all conversions and the endianness mechanism, with separate handling for register and bitfield conversions.

v2.0.17 – 2025-03-17

Features

- You can now open the enum dropdown to view the list of enum values if there is no connection open or the register is read-only.
- ModbusImport and ModbusExport now have a “-retries” parameter.
- Error codes changed:
  - Error "RESPONSE TOO SHORT AND CRC ERROR" added: shown if response was shorter than expected and has CRC mismatch. The response probably has bytes missing.
  - Error "RESPONSE FORMAT ERROR" changed to "RESPONSE PROTOCOL ERROR"
  - Error "RESPONSE DATA ERROR" added: shown for example if the query specifies to read 10 registers and less were returned is the response.

Fixes

- ModbusImport: CSV file errors were not counted when using the -verbose and -continueonerror options.
- ModbusImport and ModbusExport: Always show counters, not only in -verbose mode.
- ModbusExport: “Illegal function” errors were not counted in the error counter.
- A uint16 enum translation with a value of 65535 did not work (-1 did as workaround)

- When editing an enum value one could cause an exception with key sequences like: numlock on, start edit, press backspace to erase all, numlock off, press Enter. Or like: start edit, press backspace 3 times, press Enter.
- ModbusImport.exe and ModbusExport.exe now run as 64-bit applications.

v2.0.16 – 2023-12-27

Features

- Project configuration window: now shows a warning icon with each parameter that had a command line override, instead of one general warning.
- Command line options can now start with – or /.
- Added more command line options. Use -? -h -help /? /h or /help for help.
- Mbc file: <exportorder> element added.
- Command line import and export utilities ModbusImport.exe and ModbusExport.exe added.

Fixes

- Release notes are now in English.
- Crash fixed: an internal buffer overflow in the communications code sometimes caused an application crash.
- WriteHoldingregisterFC = ‘alternate’ is dropped because it was confusing. It is automatically replaced by ‘auto’ if found in an mbc project file.
- Stop-on-error mechanism fixed.
- Import now processes the CSV file in top-down order. Earlier versions first scanned the file for errors, then started import which could report more errors.
- Import now also ignores lines that only have spaces and/or separators.

v2.0.15 – 2023-06-24

Features

- Enum registers editing now has a dropdown combobox with possible values.
- Mbc file: Enum values now allow a custom <else> text that is shown if no value matches. <else> text can be left empty to suppress the message.
- You can now use 5 digits and 6 digits registers numbering scheme, 40001-49999 and 400001-465536, or 40000-49999 and 400000-465535, at the same time.

Fixes

- In Import_file and Import_compare: writing/comparing a float32 value no longer worked.
- Ascii32 register writing fixed.
- Resizing column width is now global for all tabs.
- Zooming now also adjusts the size of the column headers.
- You can now close the [Project_Configuration], [Save_project_file] and [About] windows with the ESC button too.
- The expand/collapse icons scale with zooming.
- If an error occurs when loading an mbc file, no auto-connect to the slave/server is done.
- With Import_file and Import_compare: writing/comparing a float32 is fixed.

v2.0.14 -- 2022-11-08

Features

- Added command line parameters. These replaces the LocalOverrides mechanism.
- LocalOverrides mechanism removed.
- Save project file: the [Update XML to current version] checkbox is now enabled by default.
- Options window: tooltips improved.
- Grid: Register column menu now has [Expand all] and [Collapse all].
- Mbc project file: <longdescr> element added. Adds a long multiline text that is show when the mouse is hoovered over the register decription.
- Mbc project file: <include> element added.

Fixes

- [Open project file location in Explorer] now works if you gave a filename without a path on the command line.
- The temporary red colouring of the value field after entering an invalid value has been improved.
- Edit enter/escape handling is improved.
- In the notification screen that the serial port could not be opened: the [Port] combo box now automatically refreshes the list of serial ports.

v2.0.13 -- 2022-07-11

Features

- Added Find/Filter bar. Open it using Control-F. There is also an option “Clear Find/Filter on tab change”.
- The properties of a register (rw, int32, ...) are now displayed in a separate [Properties] column instead of being added to the [Description] column. The [Properties] column can be enable/disabled in Options.
- The Description text can now be multi-line, if the “Multiline rows with auto row size” option is enabled. A long description text is wrapped. CR/LF’s can be used in the description text. The row height of the grid is automatically adjusted to the text.
- "Move to next row on Enter" option: depending on this option, after entering a new value with the Enter key, the cursor may or may not go to the next line. Enter a value with Control-Enter to get the opposite behaviour.

Fixes

- Control-0 for zoom reset now works globally on the window, not just when the focus is in the grid.
- Right-clicking in the grid now moves the focus to that row, to better see which row you are working on.
- ModbusControl now runs as a 64-bit application.
- Migrated to .NET framework 4.7.2.

v2.0.12 -- 2020-07-13

Features

- Added mbc project file validation with XSD schema file. Validation is optional. The validation only allows lower-case XML tags, and only allows the XML elements that are supported in this version. The intention is that over time all mbc files will migrate to meet the validation.
- In the "Save project file" window you now have the option to update the XML with "Update XML to current version" so that it then passes the validation. Errors must be corrected manually.
- In "Save project file" the project file is now saved with XML formatting.
- If you start ModbusControl by double-clicking an mbc file and the file points to a non-existent COM port, the error dialog "Error opening serial port COMx" now appears after the ModbusControl program appears.
- At the first start of the program after an update, an option is shown to see the 'Release notes'.
- Program settings split into user-dependent part and computer-dependent part. The old settings are migrated to the new settings files, then the old settings file is deleted.
- LocalOverrides.xml file is automatically moved from <mydocuments>\ModbusControl to %AppData%\Roaming\E.D.&A\ModbusControl. Content from old is not migrated. Old file is renamed with '.old' and moved to the new location. If overrides were used, the user will receive a notification.
- <user>\Documents\ModbusControl is no longer used.

Fixes

- The import/export dialogs now show '*. CSV' instead of '*. CVS'.
- Mbc file is now saved without BOM (Unicode byte order mark).
- Better handling if a non-administrator user logs on and uses ModbusControl.

v2.0.11 -- 2019-11-18

Fixes

- Better handling if the USB/serial connection is lost. In the past the program could stop responding.

v2.0.10 -- 2019-03-11

Features

- Added displaying of enum values in the value column.

Fixes

- Menu item "Edit project file" did not work correctly if there was a space in the path.
- Logger: Clear button did not clear number of logs in the status bar.
- Send Custom Modbus Command: allows tabs and CR/LF in the string. Reports error for empty command string.
- Send Custom Modbus Command: Reports error when command is too short, must be at least 2 bytes.
- Fix background colour of registers.

v2.0.9 -- 2018-10-20

Features

- New option: "Move to next row on Enter".
- New option: "Auto-start edit mode on typing".
- New option: "Use custom editor".
- New option: "Ignore form size on project reload".
- Configuration window now has an Apply button.
- Separate line-copy and value-copy capability in grid, using control-C and context menu.
- The status bar now has blue colour if there is no connection or no polling is done.
- The status bar now has red colour for errors.
- Now shows a warning in the "Save configuration to project file" and "Project configuration" screens if localoverrides were used.

Fixes

- Notification if Settings.xml could not be loaded at startup.
- Settings.xml writing now updates/inserts all tags in the existing file instead of writing a completely new file, so that any newer tags from a newer version remain.
- Project Configuration: Interframe-delay: now shows "0 (auto)" at setting 0.
- Local override project is no longer required to have a <modbus> element.
- Dialog windows can now be closed with the Enter key.
- The checkboxes of the "Save configuration to project file" window are now cleared when loading a project.
- Buttons at the bottom of the windows now have better spacing.

v2.0.8 -- 2018-08-02

Features

- Added menus and toolbar.
- Can save project configuration, bridge settings and window size in mbc file.
- Communication pausing and poll-once.
- Change the connection parameters without having to close and reopen the port.
- If the serial port is not available at start, you can immediately choose another port.
- Mbc project file: XML tags are now preferably in lowercase. The old camelcase is also still accepted.
- Always shows line numbers for errors in mbc file.
- Word order (endian) is separately configurable for int32 and float32. Character order is configurable for ASCII.
- WriteHoldingregFunctionCode now also supports 'auto'.
- User manual includes description of mbc file.